

UNIT - II

Network Layer

Network Layer Services

① Packetizing:-

The first responsibility of n/w layer is packetizing, i.e. encapsulating the payload (data received from top layer) ~~at~~ at source and decapsulating the payload at the destination.

② Routing and forwarding:-

(a) Routing:

The n/w layer is responsible for routing the pkt from its source to the destination.

A physical n/w is a combination of n/ws (LANs & WANs) and routers that connect them.

This means there are more than one route from source to destination.

n/w layer is responsible for choosing the best routes among available routes.

(b) Forwarding:

If routing is applying strategies and running some routing protocols to create the decision-making tables for each router, forwarding can be defined as the action applied by each router when a pkt arrives at one of its interface.

The decision making table is sometimes called as routing ~~table~~ & some times forwarding table.

③ Other services

- ① Error control. X
- ② Flow control X
- ③ congestion control X
- ④ Quality of service
- ⑤ Security.

Network-layer Performance.

① Delay - time incurred by pkt to reach the destination.

② Transmission delay :-

$$\text{Delay}_{tr} = (\text{pkt length}) / (\text{Transmission rate})$$

③ Propagation delay :-

$$\text{Delay}_{pg} = \frac{\text{Distance}}{\text{propagation speed.}}$$

Propagation speed in vacuum = 3×10^8 mts/sec

and it is less than the wired medium.

Propagation speed directly depends on length of link.

④ Processing delay

Delay_{pr} = Time required to process a pkt in a router or a destination host.

It's a time required by the node to receive the packet from its upper layer or port and perform error detection process & deliver to the O/P port.

(d) Queuing delay

delay_{qu} = The time a pkt waits in input and output queues in a router.

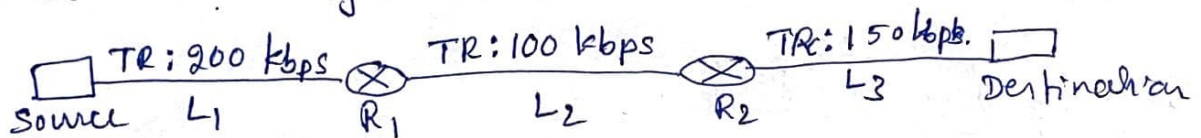
(e) Total delay

$$\text{Total delay} = (n+1)(\text{Delay}_{tr} + \text{Delay}_{pg} + \text{Delay}_{pr}) + (n)(\text{Delay}_{qu})$$

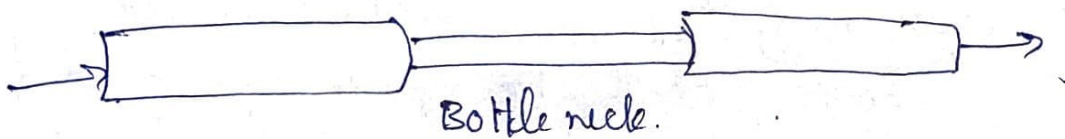
where n is the no of routers in the path from source to destination.

(2) Throughput

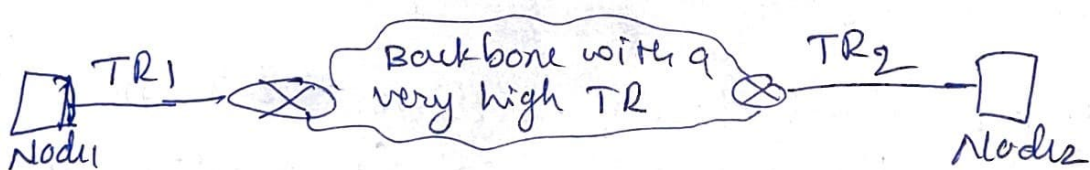
Throughput at any point in a netw is defined as the no of bits passing through the point in a ~~secs~~ second, which is actually the transmission rate of data at that point.



TR: Transmission rate

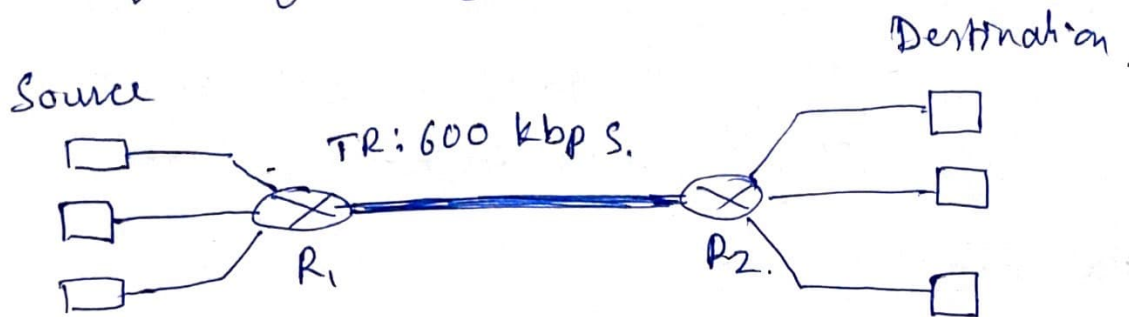


$$\text{Throughput} = \text{minimum}(TR_1, TR_2, TR_3)$$



A path through the Internet backbone.

Effect of throughput in a shared link.



Throughput of main link will be 200 kbps

4p.

③ Packet loss :

Another issue that severely affects the performance of comⁿ is the no of pkts lost during transmission. pkt loss may occur due to limited waiting buffer size at the router and receipt of pkts more than its capacity.

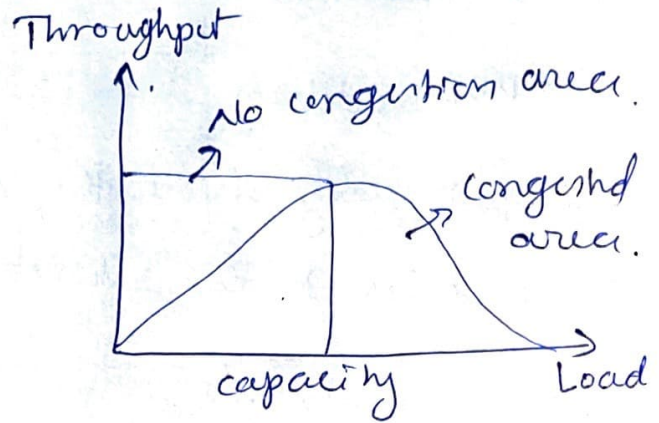
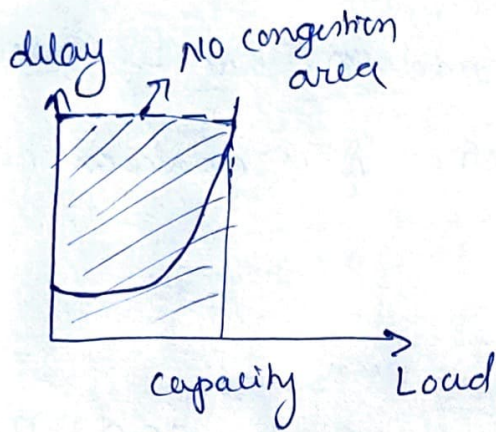
④ Congestion control :

Although congestion at the nlw layer is not explicitly addressed in the Internet model, the study of congestion at this layer may help us to better understand the cause of congestion at the transport layer and find possible remedies to be used at the nlw layer.

Congestion at nlw layer is related to two issues: delay and throughput.

delay: when load is lesser than nlw capacity the delay is mini (propagation + processing) But when load reaches the capacity, the delay increases sharply.

Throughput : If load is below n/w capacity, the throughput $\uparrow$ proportionally with load.
 But it declines sharply after reaching ~~load~~ ~~capa~~ reaches the n/w capacity.



congestion control

- ① open loop congestion control (prevention)
 - (a) Retransmission policy
 - (b) window policy (selective repeat & go-back-N)
 - (c) Acknowledgement policy
 - (d) Discarding policy
 - (e) Admission policy
- ② closed loop congestion control (removal)
 - (a) Backpressure
 - (b) choke packet
 - (c) Implicit signaling
 - (d) Explicit signaling

IPv4 Addresses

The identifier used in the ~~network~~ layer of the TCP/IP ~~protocol~~ suite to identify the connection of each device to the Internet is called the Internet address or IP address.

IPv4 address is a 32-bit address that uniquely and universally define the connection of a ~~address~~ host or a router to the Internet.

① Address spaces:

Address space is the total no of addresses used by the protocol.

If a protocol uses 'b' bits to define an address the addr space is 2^b .

IPv4 uses 32 bits, so addr. space is 2^{32}
= 4294,967,296

Notation

There are 3 common notations

Binary notation (base 2)

Dotted-decimal notation (base 256)

Hexadecimal notation (base 16).

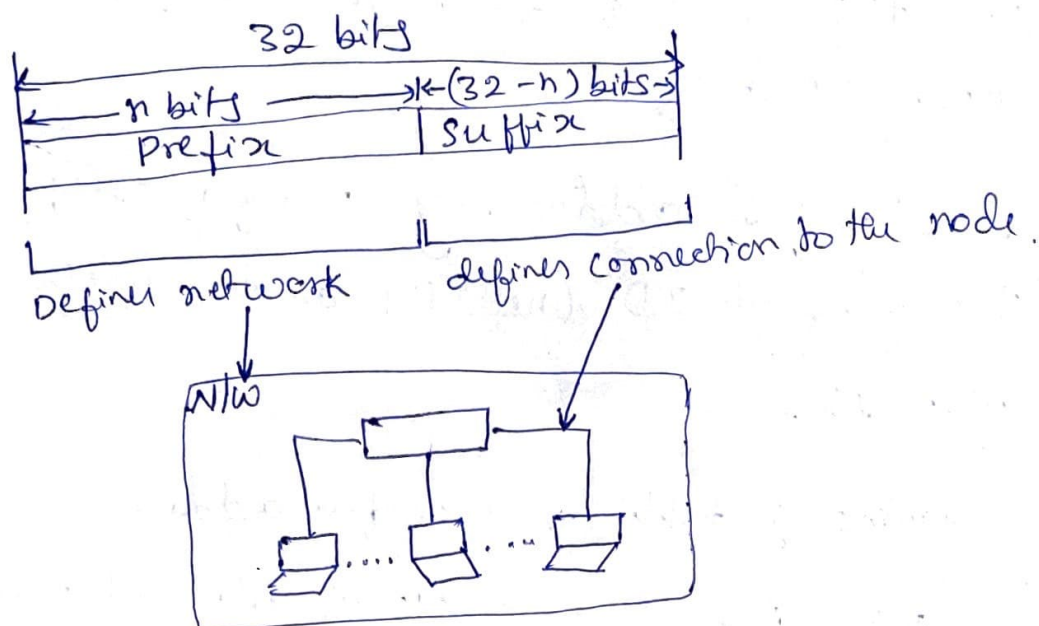
Ex:-
Binary → 10000000 00001011 00000011 00011111

128 . 11 . 3 . 31 ← Dotted-decimal

800B031F ← Hexadecimal.

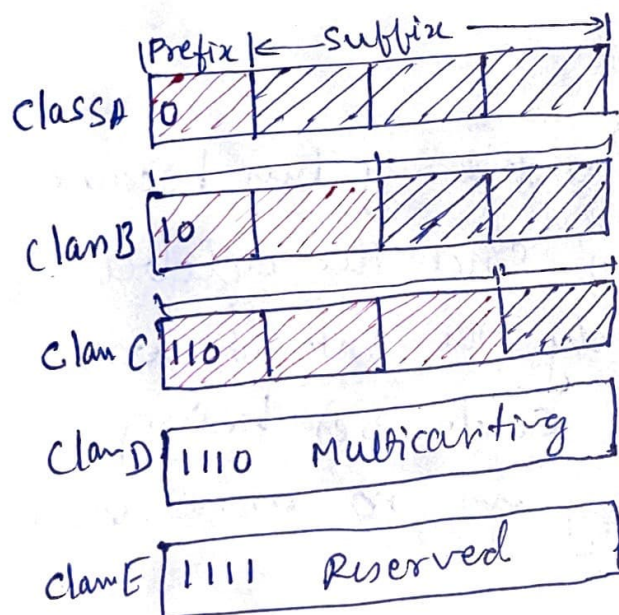
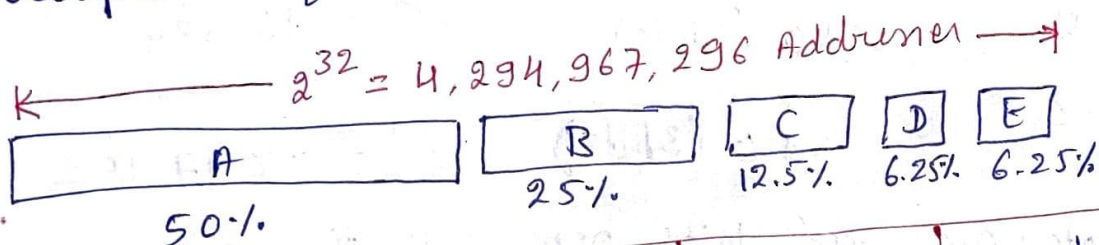
Hierarchy in addressing

IPv4 addr is hierarchical, where it is divided into 2 parts: prefix and suffix.



② Classful Addressing

Occupation of the address space in classful addressing.



Class	Prefix	First byte
A	n = 8 bits	0 to 127
B	n = 16 bits	128 to 191
C	n = 24 bits	192 to 223
D	NA	224 to 239
E	NA	240 to 255

class D is not divided into prefix and suffix.
and it is used for multicast addresses.

class E is also not divided as prefix & suffix
and this class of addresses are reserved for
future use.

In classful addressing prefix represents
the network ID (net ID) and suffix represents
host ID.

Summary table of classful addresses.

Class	Reserved bits	No. of n/w addresses in the world.
A	0 (1 bit)	$2^7 = 128$
B	10 (2 bits)	$2^{14} = 16,384$
C	110 (3 bits)	$2^{21} = 2,097,152$

Note: reserved bits represents the class.

Address depletion:

The reason that classful addressing has become
obsolete is address depletion. Since the addresses
were not distributed properly, the Internet was
faced with the problem of the addresses being
rapidly used up, resulting in no more addrs.

available for organizations and individuals that needed to be connected to the Internet.

Subnetting and Supernetting

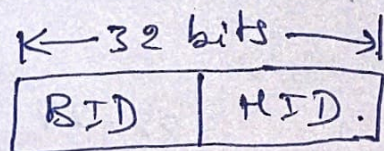
To alleviate address depletion, two strategies were proposed and, to some extent, implemented: Subnetting and Supernetting.

In Subnetting, a class A or class B block is divided into several subnets.

While Subnetting was devised to divide a large block into smaller ones, Supernetting was devised to combine several class C blocks into a larger block to be attractive to the organization. However, routing was difficult in this method.

Classless Addressing

- * No class concept here, but only concerned with blocks.
- * IP addresses advantage is reduced



BID - Block ID

Host-ID - Host-ID

* Notation:

byte . byte . byte . byte / n.

no. of bits
used to represent
net id.

Ex: 12.24.76.8 / 8

23.14.67.92 / 25 etc

Rules :-

- ① Address should be contiguous
- ② No of addresses in a block must be in power of 2
- ③ First address of every block must be evenly divisible with size of block

Extracting information from an address

Given any address in the block, three piece of infoⁿ about the block to which the address belongs :

- ① No of addresses
- ② First address in the block.
- ③ Last address in the block.

can be identified as follows.

- ① No of address in the block $N = 2^{32-n}$
- ② To find the first address, keep n leftmost bits and set the $(32-n)$ rightmost bits all to 0's.
- ③ To find the last address, keep n leftmost bits and set the $(32-n)$ rightmost bits all to 1's.

Ex:- Given address 167. 199. 170. 82/27.

Here $n = 27$.

- ① No of addresses $N = 2^{32-27}$
 $= 2^5 = 32$ addresses

27 bits	5 bits
BID	MSD

N is the size of the block.

② The first address can be found by keeping the first 27 bits & changing rest of the bits to 0's.

10100111 11000111 10101010 01010010
167 . 199 . 170 . 82

we get the following after replacing ~~rest~~ 5 bits with 0's.

10100111 11000111 10101010 01000000
167 . 199 . 170 . 64

Finally, the first address of the block is

$167.199.170.64/27$

② The last address can be found as follows.

10100111 11000111 10101010 01010010
167 . 199 . 170 . 82

we get the following address after replacing 5 LSB with 1's

10100111 11000111 10101010 01011111
167 . 199 . 170 . 95

Finally, the last address of the block is

$167.199.170.95/27$

32 addresses of the block is ranging from.

$167.199.170.64/27$, to $167.199.170.95/27$

Address mask

Another way to find and last addresses in the block is to use the address mask. The address mask is a 32-bit no in which the leftmost bits are set to 1s and the rest of the bits $(32-n)$ are set to 0s.

A computer can easily find the address mask because it is a complement of $(2^{32-n} - 1)$

The information in the block can be extracted using 3 bit-wise operations NOT, AND and OR as follows.

1. The no of addresses in the block $N = \text{NOT}(\text{mask}) + 1$.
2. The first address in the block = (Any address in the block) AND (mask)
3. The last address in the block = (Any address in the block) OR (NOT (mask))

Ex:- Given address 167. 199. 170. 82 / 27.

mask is 255. 255. 255. 224

$$\textcircled{1} N = \text{NOT}(\text{mask}) + 1 = 0. 0. 0. 31 + 1 = 32 \text{ addresses}$$

$$\textcircled{2} \text{ First address} = \begin{array}{r} 167. 199. 170. 82 \\ \text{AND} \quad 255. 255. 255. 224 \\ \hline 167. 199. 170. 64 / 27 \end{array}$$

$$\textcircled{3} \text{ Last address} = \begin{array}{r} 167. 199. 170. 82 \\ \text{OR} \quad 0. 0. 0. 31 \\ \hline 167. 199. 170. 95 / 27 \end{array}$$